



REAL-TIME INTENT-AWARE E-COMMERCE PERSONALIZATION USING AI

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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LIST OF ABBREVIATION

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CERTIFICATE

ABSTRACT

The rapid growth of e-commerce platforms has transformed the way people shop and interact with online marketplaces. With thousands of products available on digital platforms, users often find it difficult to identify items that match their interests and preferences efficiently. Traditional recommendation systems used in many e-commerce websites mainly rely on static approaches such as collaborative filtering, popularity-based suggestions, and historical purchase data. Although these methods provide basic recommendations, they fail to understand the real-time intent of users during their browsing sessions. As a result, users may receive irrelevant product suggestions, which reduces engagement and negatively affects the overall shopping experience.

The proposed system, **Real-Time Intent-Aware E-Commerce Personalization using Artificial Intelligence**, aims to improve the effectiveness of recommendation systems by dynamically identifying user intent and providing personalized product suggestions. The system continuously monitors user activities such as search queries, product clicks, browsing history, time spent on specific product pages, and cart interactions. These behavioral patterns are analyzed using artificial intelligence and machine learning techniques to determine the user's current intent, which may include browsing products, comparing multiple items, or preparing to make a purchase.

Based on the detected intent, the system generates personalized recommendations, customized offers, and adaptive product displays in real time. This approach enables the platform to provide more relevant product suggestions and improves the efficiency of product discovery. The use of machine learning models allows the system to adapt to changing user behavior and provide intelligent recommendations within the same browsing session.

The proposed solution not only enhances user satisfaction but also benefits e-commerce businesses by increasing customer engagement, improving conversion rates, and optimizing marketing strategies. Furthermore, the system is designed to be scalable and capable of handling large volumes of user data, making it suitable for integration into modern e-commerce platforms.

Overall, the implementation of real-time intent-aware personalization demonstrates how artificial intelligence can significantly enhance the performance and effectiveness of recommendation systems in online shopping environments, creating a more interactive and user-centric digital marketplace.